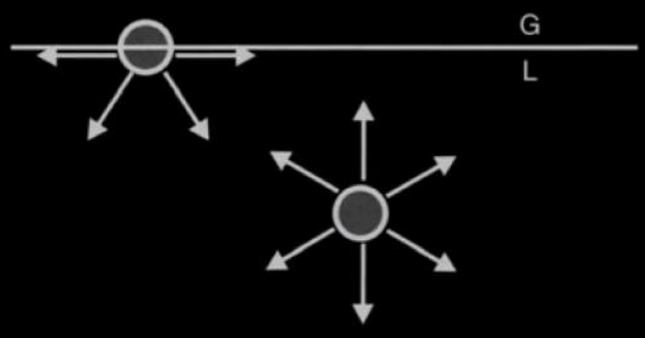
Origin of Surface Energy:



Cohesive Forces: Attractive forces between same type of molecules are known as cohesive forces.

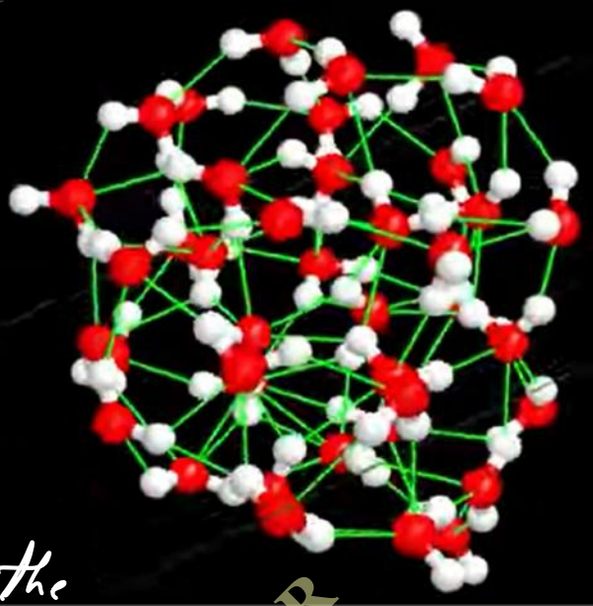
When a molecule is well within the volume of a liquid, it is symmetrically surrounded by molecules from all sides however near the surface this symmetry is broken, as clear from the figure, there is a net downward pull on a surface molecule and for a particle to reach the surface, a negative work is done by these forces causing a higher potential energy at the surface.

∴ All liquids tend to attain least possible surface area (∵ drops attain a spherical shape ∵ a sphere has least surface area for a given volume).

Whenever surface area of a liquid is to be increased, the molecules on the existing surface must move away from each other against the cohesive pull, thus the work done by these cohesive forces is also negative and causes an increase in surface energy.

→ Since molecules on surface can't form bonds with any molecules from above, they form stronger bonds with surface molecules thus causing a large force per unit length on any line drawn on surface.

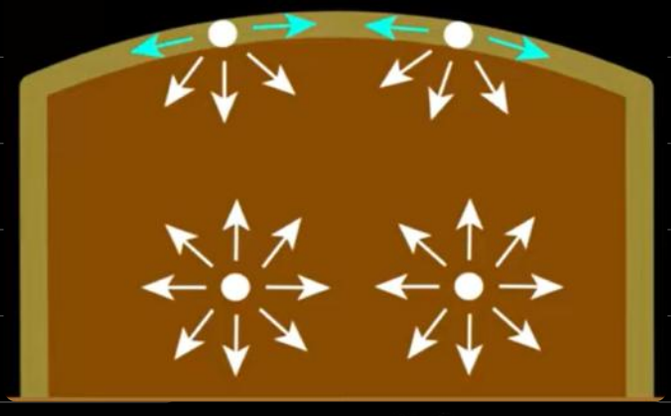
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For instance consides a water drop. There is nobody to hold the hand of $O^{\delta-}$ on top similarly there is nobody to hold the hand of $H^{\delta+}$ from abov so they decide to "Hold" each others hands. If the drop is forced to enpand then the surface molecules will need to mov away from each other thus causing an increase in potential energy (∵ negativ work done by cohesiv forces on the surface).

This stronger bonding on surface causes the surface to behav like a stretched membrane (like say rubber membrane on a balloon surface)

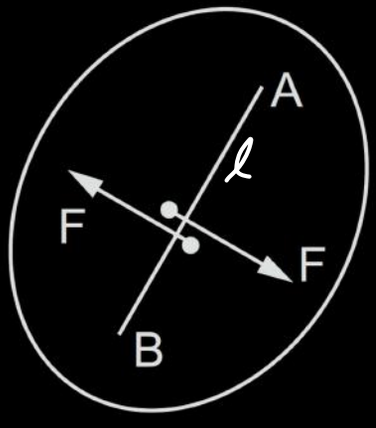
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Surface Tension: Force per unit length acting on a line drawn over a surface of a liquid is called surface tension.

For instance consider a line AB on surface of water. The molecules on either side of the line attract each other with a force F (say) then

$$T = F/l$$



Effect of Temperature on Surface Tension: Increase in temperature leads to an increase in random thermal motion thus reducing the effective bonding forces and causing a reduction of surface tension.

Effect of adding highly soluble Salt (like NaCl) on surface tension of water: NaCl breaks in to Na^+ and Cl^- thus forming stronger bonds with $H^{\delta+}$ and $O^{\delta-}$ thus increasing the surface tension.

Effect of adding detergents: Detergents have a hydrophilic part which loves to stay deep away from surface and well surrounded by water molecules and the hydrophobic part stays on surface (to avoid water molecules as much as possible) Thus reducing the surface tension.)

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Example:

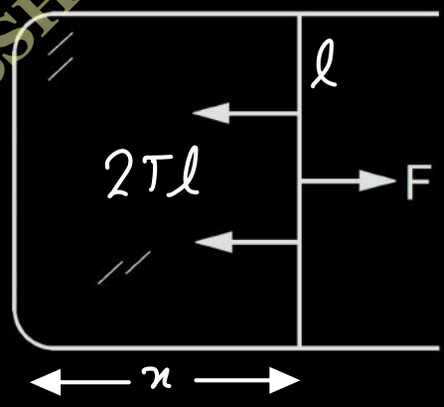
Water is kept in a beaker of radius 5·0 cm. Consider a diameter of the beaker on the surface of the water. Find the force by which the surface on one side of the diameter pulls the surface on the other side. Surface tension of water = 0·075 N m^{-1}.

Sol: $D = 2R = 10\text{ cm} = 0.1\text{ m}$, Now $T = F/l$

$\Rightarrow F = Tl = 0.075\text{ N/m} \times 0.1\text{ m} = 7.5 \times 10^{-3}\text{ N}.$

Q. Suppose instead of the water surface on the beaker, it were a film of water (say in a bangle), What would be the surface tension force applied by film on one side of the diameter on the other side?

Ans: ∵ The film has two surfaces, surface tension forces would be acting on both liquid air interfaces and as a result the surface tension force would be double i.e. 15×10^{-3} N.



Relation Between Surface Tension and Surface Energy: Consider the U shaped frame with a slider stretching a soap film. If we pull the slider slowly through a distance x, the work done by external agent in the process must be $W = Fx = 2Tlx$. This must manifest as increase in surface energy. $\Rightarrow U = 2Tlx$.

Now let us find surface energy per unit surface Area (i.e. the total area of liquid-air interface). $A = 2lx \Rightarrow u = \frac{U}{A} = \frac{2Tlx}{2lx}$

$\Rightarrow \boxed{u = T}$ This gives us an alternative definition of surface tension as surface energy per unit surface area.

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Example:

A water drop of radius 10^{-2} m is broken into 1000 equal droplets. Calculate the gain in surface energy. Surface tension of water is $0{\cdot}075\ \mathrm{N\,m^{-1}}$.

Sol: By volume conservation: $\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi r^3 \Rightarrow r = \frac{R}{10}$

where $R \rightarrow$ radius of original drop, $r \rightarrow$ radius of smaller drops.

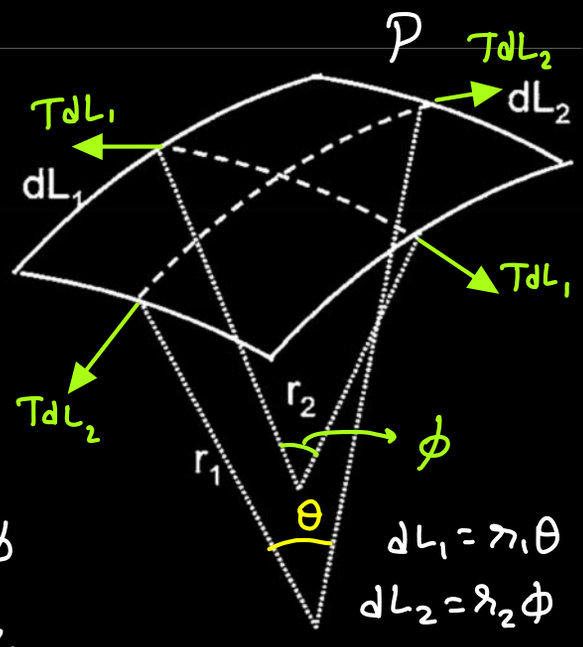
Now gain in surface energy $= \Delta U = T\left[1000 \times 4\pi r^2 - 4\pi R^2\right]$

$\Rightarrow \quad \Delta U = 36\pi R^2 T = 8{\cdot}5 \times 10^{-4}$ Joules.

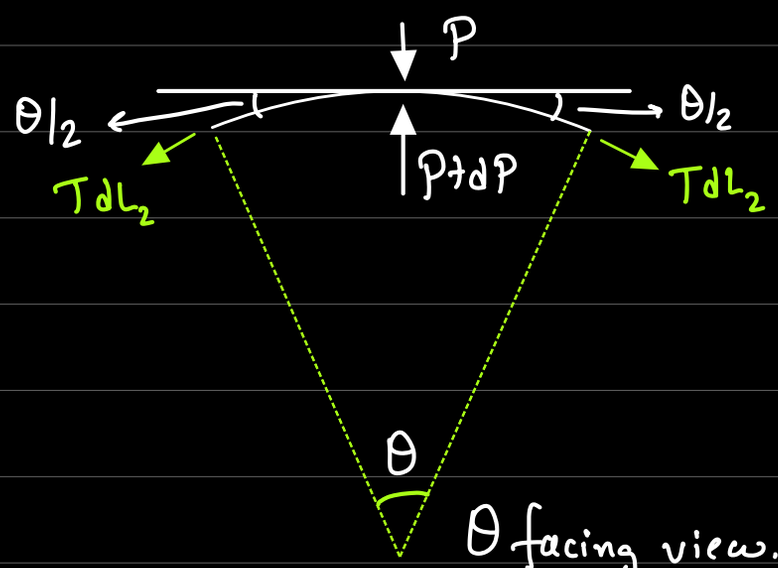
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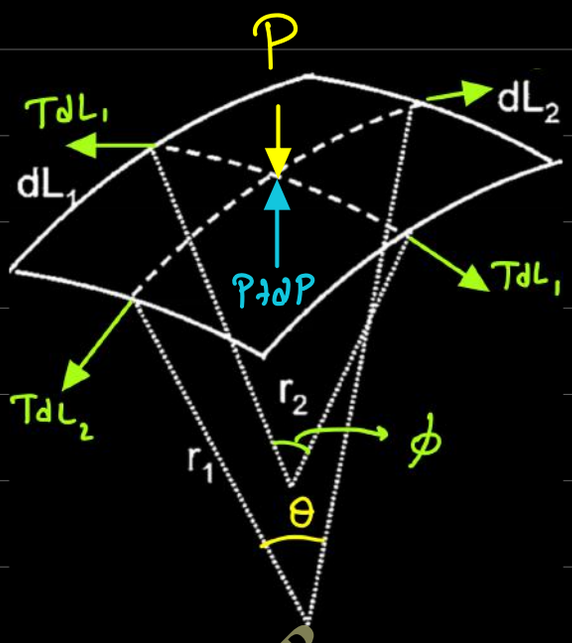
Young Laplace Equation. (Excess pressure across a liquid air interface).

Consider small patch of liquid air interface with two mutually ⊥ radii r_1 and r_2 with film subtending corresponding angles θ & ϕ at the respective centres of curvature.



The surface tension forces acting on this patch are pulling it downward and these must be balanced by excess pressure on the concave side.





From θ facing view, visible downward surface tension force:

$F_1 = 2T\,dL_2 \sin(\theta/2) = T\,dL_2\,\theta$ ($\because \theta$ is very small)

Similarly from ϕ facing view: $F_2 = 2T\,dL_1 \sin(\phi/2) = T\,dL_1\,\phi$

Also net upward pressure force: $F = \Delta P \times dL_1 \times dL_2$

Now by Force Balance, $F = F_1 + F_2$

$$\Rightarrow \quad T\,dL_1\,\phi + T\,dL_2\,\theta = \Delta P\,dL_1\,dL_2$$

$$\Rightarrow \quad \Delta P = \frac{T\phi}{dL_2} + \frac{T\theta}{dL_1}, \quad \text{Also: } dL_1 = r_1\theta,\ dL_2 = r_2\phi$$

$$\Rightarrow \quad \boxed{\Delta P = \frac{T}{r_1} + \frac{T}{r_2}} \leftarrow \text{Young Laplace Equation.}$$

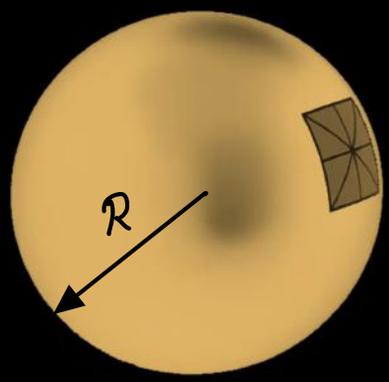
Excess Pressure in a Liquid Drop:

For the shown liquid Drop:

$r_1 = r_2 = R$

$$\Rightarrow \quad \Delta P = T/R + T/R = 2T/R$$

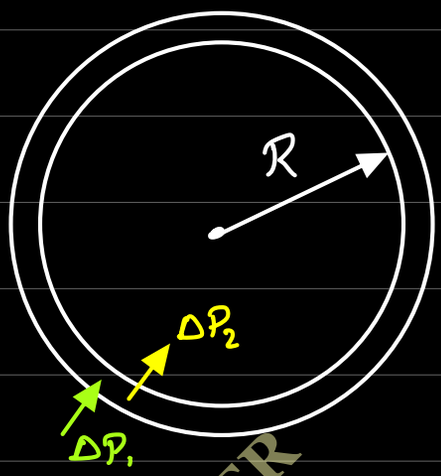
$$\Rightarrow \quad \boxed{\Delta P = 2T/R}$$



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Excess Pressure in a Soap Bubble:

For reaching the inside of a soap bubble, one has to cross two liquid air interfaces ∴ YL equation has to be applied twice.

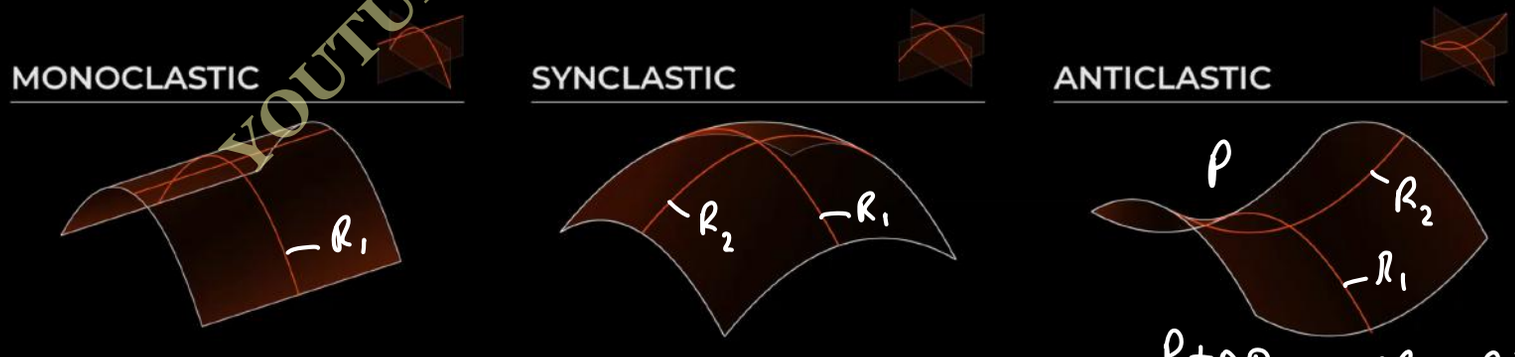


$\Rightarrow \Delta P = (T/R + T/R) + (T/R + T/R) = 4T/R$

$\Rightarrow \boxed{\Delta P = 4T/R}$

Similarly we can also calculate excess pressure for a bubble in aquarium. ∴ there will be only one liquid air interface for an aquarium bubble $\Rightarrow \boxed{\Delta P = 2T/R}$

Young Laplace Equation for anticlastic Surfaces:



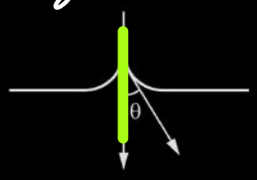
$\Delta P = \frac{T}{R_1}$

$\Delta P = \frac{T}{R_1} + \frac{T}{R_2}$

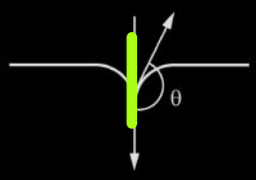
$\Delta P = \frac{T}{R_1} - \frac{T}{R_2}$

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Angle of Contact: When a glass plate (or any other material) is inserted in a liquid, the angle between the solid liquid contact plane and the liquid gas contact tangent plane is called angle of contact.



Glass in water



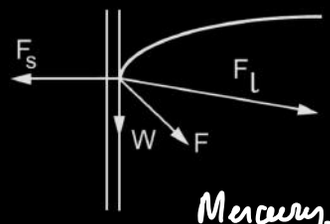
Glass in Mercury.

The Cause of Angle of Contact: In steady state, a liquid can't bear any shearing stress. ∴ the surface of a liquid assumes a shape perpendicular to the resultant of adhesive and cohesive forces

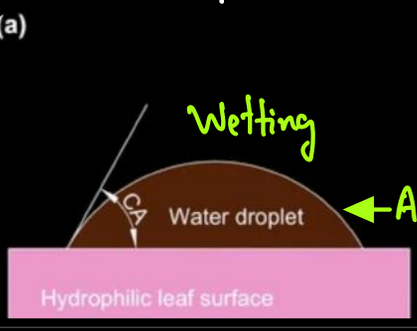
For instance when water is filled in a glass beaker, the adhesive forces are comparatively larger than cohesive forces leading to the resultant being outside the vessel thus leading to a concave meniscus. In contrast, in mercury the cohesive forces dominate thus making the resultant of cohesive and adhesive forces inside the vessel causing a convex meniscus. This also explains the shape of liquid drops on solid surfaces.



Water



Mercury.



Acute Angle of Contact

Adhesive Forces Dominate



Obtuse Contact.

Cohesive Forces Dominate.

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Capillary action: When a fine capillary made of some material is inserted in a liquid it is found that the surface of liquid in the capillary either rises or falls below the free surface of liquid in the vessel. (This happens due to surface tension effects).

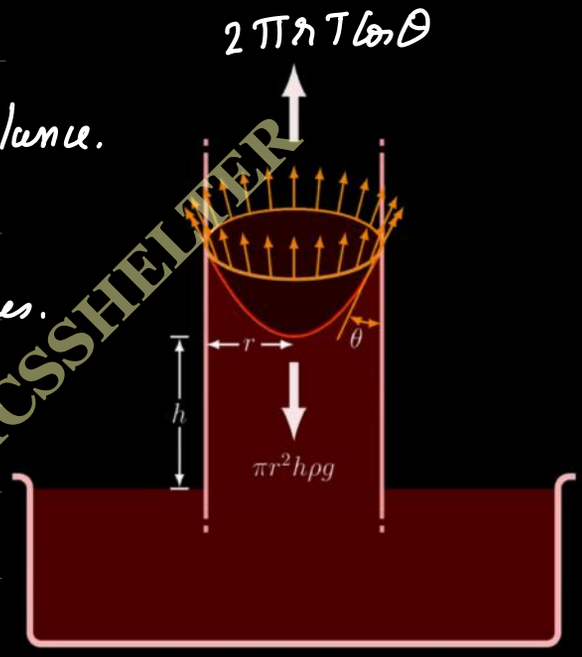
Calculating Capillary Rise: Method I: Force Balance.

The weight of the risen liquid is balanced by the surface tension forces.

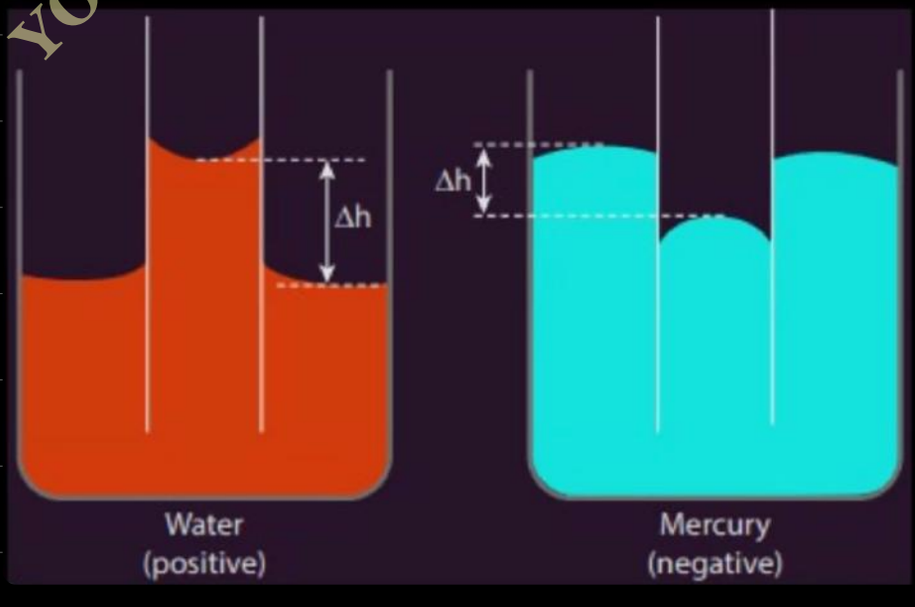
$\Rightarrow 2\pi r T\cos\theta = \rho\pi r^2 h g$

$\Rightarrow$

$$\boxed{h = \frac{2T\cos\theta}{r\rho g}}$$



If θ is obtuse, h comes out to be negative, which means, instead of a capillary rise, there will be a capillary fall.

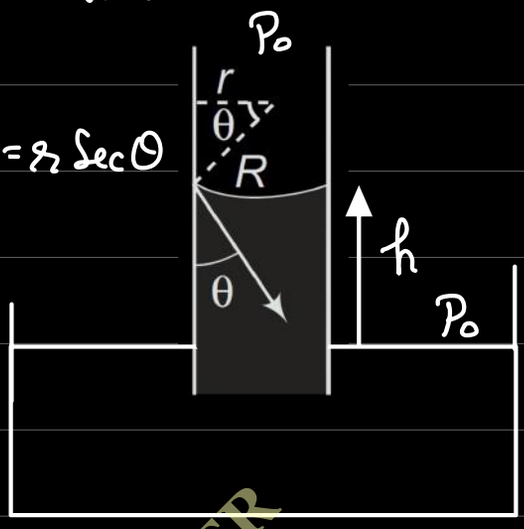


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Method II: Using Excess pressure on concave side:

$$P_0 - \frac{2T}{r\sec\theta} + \rho g h = P_0$$

$R = r\sec\theta$

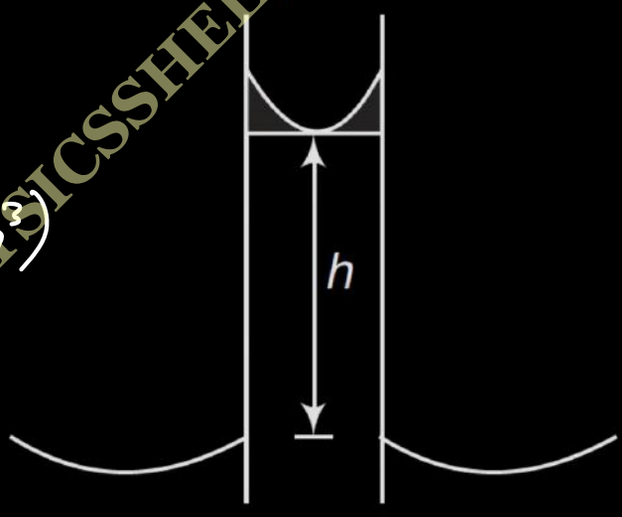


$$\Rightarrow \boxed{h = \frac{2T\cos\theta}{\rho g r}} \quad (I)$$

Correcting for hemispherical shape of meniscus ($\theta = 0°$, water in glass):

$$T \times 2\pi r = \rho g \left(\pi r^2 (h + r) - \frac{2}{3}\pi r^3\right)$$

$$\Rightarrow \boxed{h = \frac{2T}{\rho g r} - \frac{r}{3}}$$

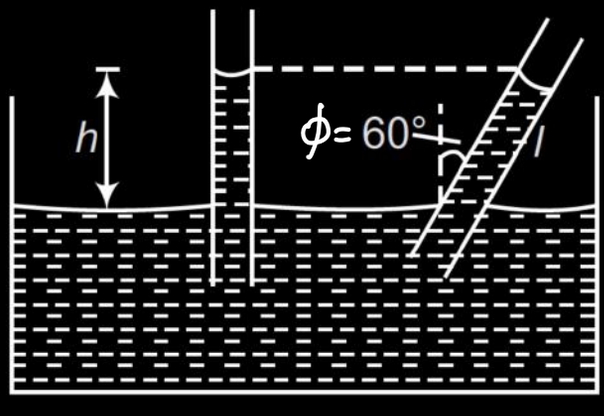


Inclined Capillary:

Balancing forces along inclined length of the capillary:

$$2\pi r T\cos\theta = \rho \pi r^2 l g \cos\phi$$

$$\Rightarrow l\cos\phi = h = \frac{2T\cos\theta}{r\rho g}$$



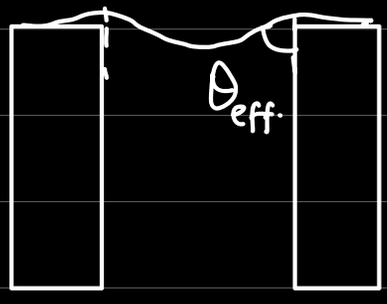
$\Rightarrow$ The vertical projection of l in the capillary is same as the risen height in the vertical capillary.

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Capillary of insufficient length: When the length of a capillary is not sufficient as required by the equation (I), the liquid rises all the way to the top of the capillary and the effective angle of contact adjusts so that equation (I) is true with this adjusted angle. [In case of capillary fall, the fall will occur all the way to the bottom of capillary of insufficient depth).

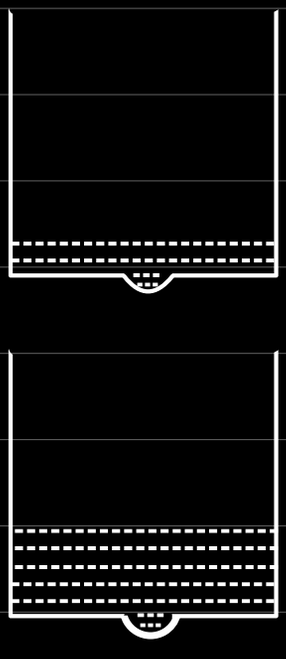
The capillary overflow can't occur steadily because it would lead to violation of energy conservation. (Free pumping of liquid from a lower level to a higher level).

Note that water will sit on top of the tube and at that point angle of contact will be same as usual.



Maximum Water level in a vessel with a hole:

As more and more water is filled in a vessel with hole, the water bulges out of the hole such that excess pressure across the bulge is exactly equal to ρgh i.e $2T/R = \rho gh$. Thus as more and more water is filled, the bulge increases gradually to a hemispherical shape at which its radius of curvature is minimum, any bulging beyond this will cause



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a leakage of water (as further bulging increases the radius of curvature thus reducing the pressure inside the bulge and now static equilibrium is not possible anymore).

Thus at maximum height:

$$\rho g h = 2T/r \quad \text{where } r \text{ is the hole radius}$$

$$\Rightarrow \boxed{h = 2T/\rho g r}$$

A similar result can be obtained for a vessel with a hole being pushed in to a sink without water filling the vessel.

Maximum Allowable column length in a vertical capillary:

For supporting maximum column length, the lower bulge must be hemispherical in shape as seen above and the upper meniscus will be governed by contact angle.

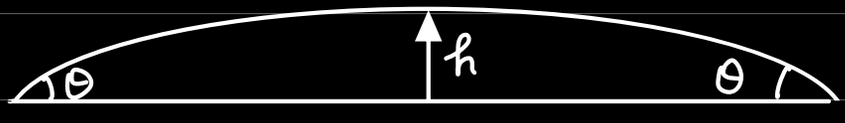


Now by force balance:

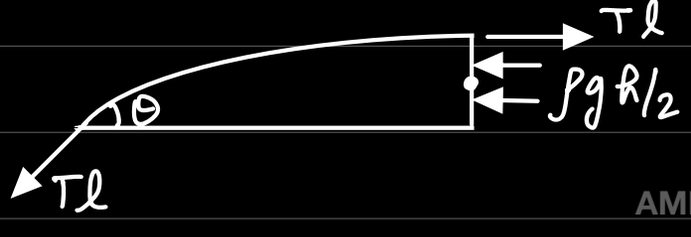
$$2\pi r T \cos\theta + 2\pi r T = \rho \pi r^2 h g.$$

$$\Rightarrow \boxed{h = \frac{2T(1+\cos\theta)}{\rho g r}}$$

Height of a Puddle:



Consider the FBD of a unit depth of puddle excluding the bottom



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layer of water molecules. By horizontal force balance:

$$Tl = Tl\cos\theta + \rho g \frac{h}{2}(lh)$$

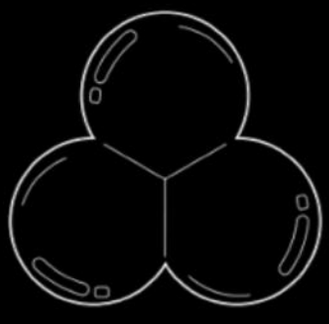
Rearranging we get:

$$\boxed{h = \sqrt{\frac{2T(1-\cos\theta)}{\rho g}}}$$

Heat released in Capillary Rise: When water rises in a capillary, it gains kinetic energy due to work done by surface tension forces, ∴ the level oscillates about the equilibrium position and finally settles due to damping. The heat released in the process can be readily found by subtracting gain in PE of the liquid from the work done by surface tension forces.

Interface of three films:

Whenever three films meet in a line, the films must be at an angle of 120° from each other because the three films intersect in a line exerting equal forces per unit length at the line of contact and the only way equilibrium can be established using three forces of equal magnitude is by putting them at 120° each.

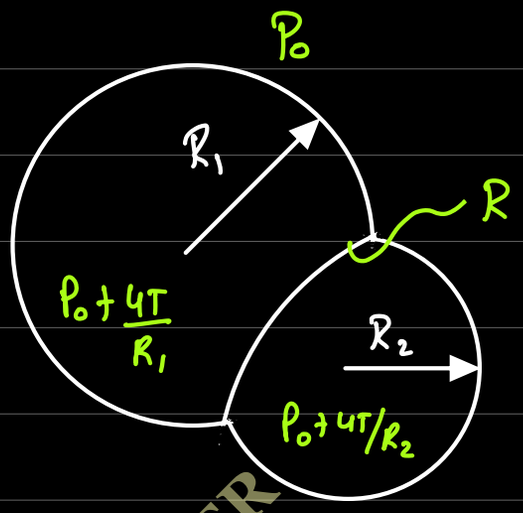




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Example: Internal surface of a double Bubble.

Find the radius of curvature of internal film of a double bubble shown.



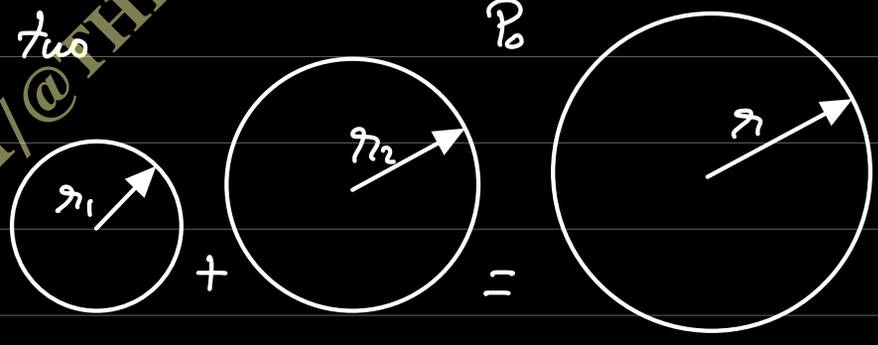
Sol: Using Concept of excess pressure (concave side at higher pressure)

$$P_0 + \frac{4T}{R_1} + \frac{4T}{R} = P_0 + \frac{4T}{R_2}$$

$$\Rightarrow \quad \boxed{R = \frac{R_1 R_2}{R_1 - R_2}}$$

Example: (Combining Bubbles (mole Conservation)

Under isothermal conditions, two bubbles coalesce to form a single bubble as shown. Find the surface tension if atmospheric pressure is P_0.



Sol: By mole conservation:

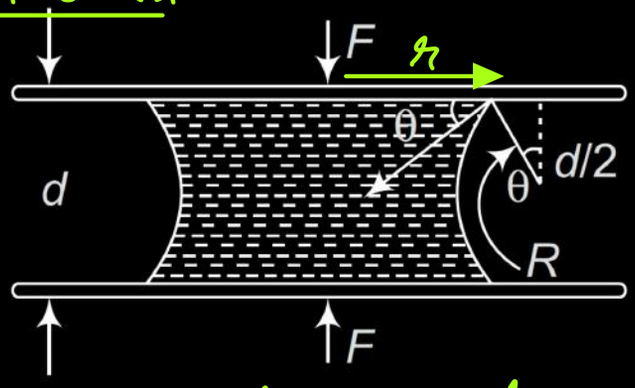
$$\left(P_0 + \frac{4T}{r_1}\right) \times 4/3\,\pi r_1^3 \times \frac{1}{\theta} + \left(P_0 + \frac{4T}{r_2}\right) \frac{4}{3}\pi r_2^3 \times \frac{1}{\theta}$$

$$= \left(P_0 + \frac{4T}{r}\right) \times \frac{4}{3}\pi r^3 \times \frac{1}{\theta} \qquad (\theta \rightarrow \text{Temperature})$$

Rearranging and solving we get: $T = \dfrac{P_0 (r^3 - r_1^3 - r_2^3)}{4(r_1^2 + r_2^2 - r^2)}$

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Force required to Separate Microscope Slides:



The surface of water between microscope slides forms an anticlastic profile as shown. Due to small value of d, there is pressure deficit in the region containing water.

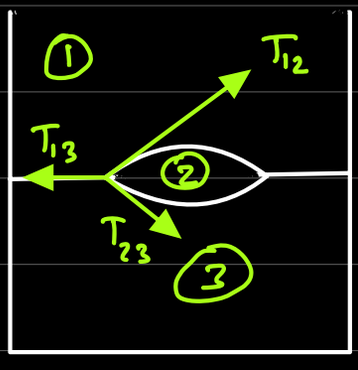
Now by Y-L equation:

$$\Delta P = \frac{T}{d/2\ \sec\theta} - \frac{T}{R} \quad \left(\approx \frac{2T}{d\sec\theta} \because d << R\right)$$

Now $F = \Delta P \times \pi R^2$

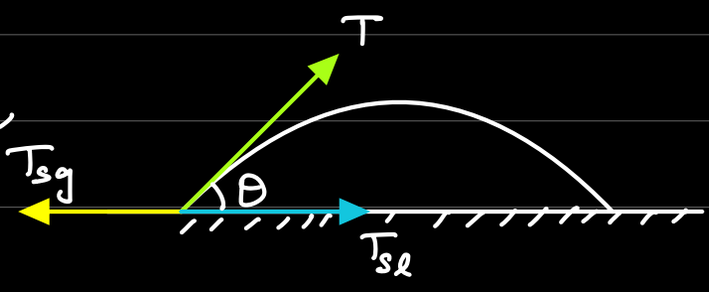
Interfacial Tension: Just like surface tension at the liquid gas boundary, there is a pulling force per unit length at solid liquid or solid gas interface, this is known as Interfacial Tension.

Young Dupre' Equation: When a contour is touching three fluid media (say an oil drop on surface of milk) then $\vec{T}_{12} + \vec{T}_{23} + \vec{T}_{13} = 0$. For solids this force balance is applied parallel to solid surface (i.e in the direction where movement of contour is possible).



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Understanding Angles of Contact and Wetting in terms of interfacial tensions:



For equilibrium along surface,

$T\cos\theta + T_{sl} = T_{sg}$

$\Rightarrow \cos\theta = \left(\frac{T_{sg} - T_{sl}}{T}\right)$ (I)

Thus if $T_{sg} > T_{sl}$, we get an acute angle of contact and we say that it is a wetting liquid ($\theta < \pi/2$) (We can also think of this in terms of surface energy, since $T_{sg} > T_{sl} \Rightarrow$ system can reduce its potential energy by increasing s-l interface and decreasing s-g interface). Also if $T_{sg} - T_{sl} > T$ then acc eqn (I), $\cos\theta > 1$ which is not possible, in such a situation, the liquid keeps spreading indefinitely (Theoretically until it is a single molecule thick layer). Such a liquid is said to be perfectly wetting liquid.

Similarly if $T_{sg} < T_{sl}$, the liquid can reduce its PE by reducing s-l contact and increasing s-g contact. In this case the contact angle is obtuse and the liquid is said to be non-wetting. Further if $T_{sl} - T_{sg} < T$, $\cos\theta < -1$ acc eqn (I), in this case the s-l interface reduces to a single point theoretically and the liquid is said to be perfectly non-wetting.

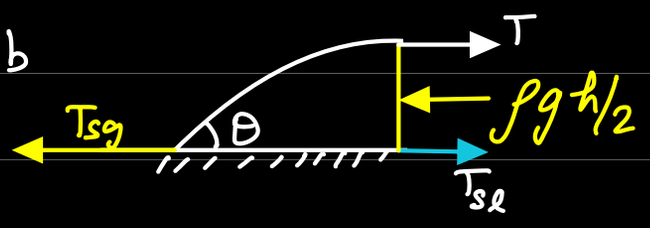
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Height of a puddle using interfacial tensions:

Let the inward dimension of Puddle be b

then :

$$T_{sg} \times b + \rho g \frac{h}{2} \times hb = T_{sl} \times b + T \times b$$



$$\Rightarrow h = \sqrt{\frac{2(T_{sl} - T_{sg} + T)}{\rho g}}$$

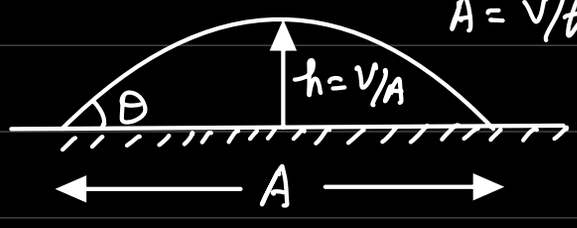
[Note FBD included last liquid molecule therefore T_{lg} is not shown]

$$\Rightarrow h = \sqrt{\frac{2T(1-\cos\theta)}{\rho g}}$$

This is the same result that we obtained earlier.

Height of a Puddle using Energy Approach: The shape of the puddle will be such that the total Potential Energy of the system (Surface Energy + Gravitational PE) is minimized.

Now when the puddle spreads, the solid gas interface decreases and solid liquid and liquid gas interfaces increase. Also gravitational PE decreases.



Now let a volume V spread to an area A.

$$\Rightarrow U = \rho V g \frac{h}{2} + T\left(\frac{V}{h}\right) + T_{sl}\left(\frac{V}{h}\right) - T_{sg}\left(\frac{V}{h}\right)$$

Now $\frac{dU}{dh} = 0$ for least PE.

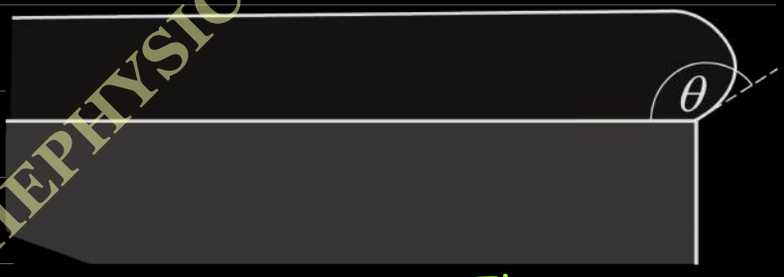
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$$\Rightarrow \frac{\rho g}{2} - \frac{T}{h^2} - \frac{T_{sl}}{h^2} + \frac{T_{sg}}{h^2} = 0$$

$$\Rightarrow h = \sqrt{\frac{2(T + T_{sl} - T_{sg})}{\rho g}} = \sqrt{\frac{2T(1-\cos\theta)}{\rho g}}$$

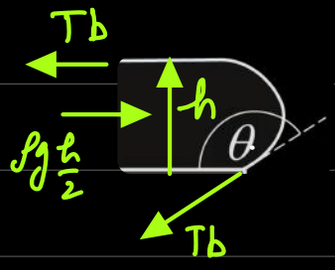
Example. PATHFINDER BUILD-8

An amount of water is poured on top circular face of a horizontal glass cylinder. Excess amount of water spills over the edges and a little amount of water remains there in a thin uniform layer, a portion of which is shown in the figure. If contact angle is θ and surface tension of water is T, determine thickness of the water layer left.



Sol: Consider the FBD of a portion of the liquid layer with inward depth b.

(We have left one molecule thick layer at the bottom so as to be able to work only in terms of surface tension. Now balancing horizontal forces:



$$Tb + Tb\cos(180^\circ - \theta) = \rho g \frac{h}{2} \times hb$$

$$\Rightarrow h = \sqrt{\frac{2T(1-\cos\theta)}{\rho g}} \quad \text{Ans.}$$

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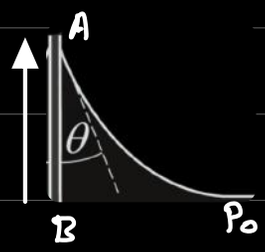
Example: PATHFINDER CHECK-4:

When a glass plate is inserted vertically in water, water rises along the plate making its surface close to the plate curved as shown in the figure. Find expression for the height to which water rises along the plate. Surface tension of water is T, density of water is ρ and acceleration due to gravity is g.

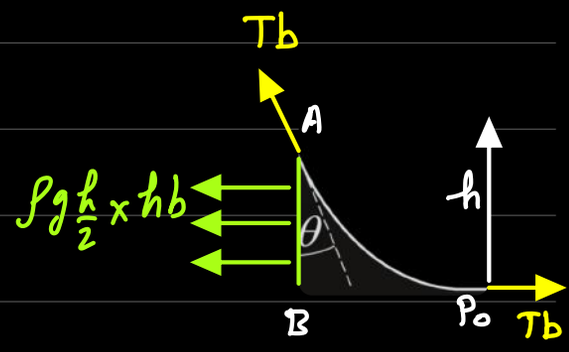


Solution: Consider FBD of the portion of water above the free horizontal level of liquid to the right of the plate. We shall work in terms of gauge pressures.

The gauge pressure at the section AB must be negative because at horizontal level it is P_0 and pressure decreases with height. Thus pressure at centroid of AB is $-\rho g h/2$ or a suction pressure of $\rho g h/2$.



By force balance:



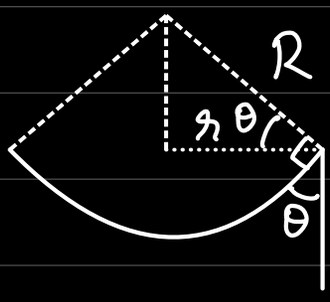
$$Tb\sin\theta + \rho g\frac{h}{2}hb = Tb$$

$$\Rightarrow \quad h = \sqrt{\frac{2T(1-\sin\theta)}{\rho g}}$$

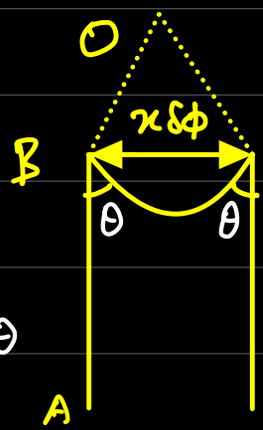
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Example : Irodov : 2.170 :

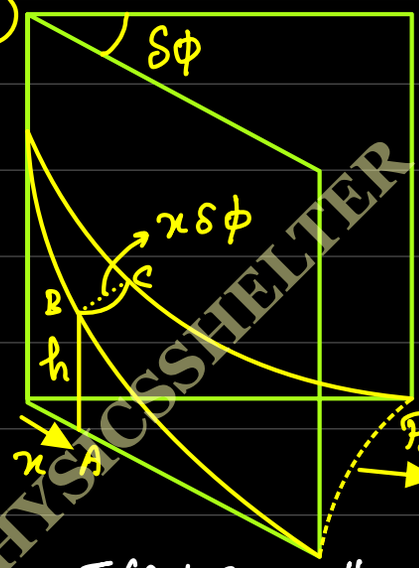
Two vertical plates submerged partially in a wetting liquid form a wedge with a very small angle $\delta\phi$. The edge of this wedge is vertical. The density of the liquid is ρ, its surface tension is α, the contact angle is θ. Find the height h, to which the liquid rises, as a function of the distance x from the edge.



$$R = x \sec\theta = \frac{x\,\delta\phi}{2}\sec\theta$$



Front View :



Using Young Laplace Equation (Considering Effect of smaller radius alone)

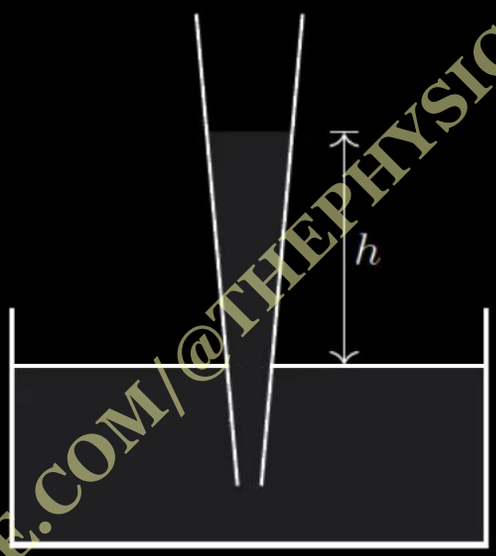
$$P_0 - \frac{T}{R} = P_0 - \rho g h \quad \Rightarrow \quad h = \frac{T}{\rho g R}$$

$$\Rightarrow \quad h = \frac{2T\cos\theta}{\rho g x \Delta\phi} \quad \text{Ans}$$

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Example: JEE Advanced: 2014:

Q 1. A glass capillary tube is of the shape of a truncated cone with an apex angle α so that its two ends have cross sections of different radii. When dipped in water vertically, water rises in it to a height h, where the radius of its cross-section is b. If the surface tension of water is S, its density is ρ, and its contact angle with glass is θ, the value of h will be (where g is acceleration due to gravity) (2014)



(A) $\frac{2S}{b\rho g}\cos(\theta - \alpha)$ (B) $\frac{2S}{b\rho g}\cos(\theta + \alpha)$

(C) $\frac{2S}{b\rho g}\cos(\theta - \alpha/2)$ (D) $\frac{2S}{b\rho g}\cos(\theta + \alpha/2)$

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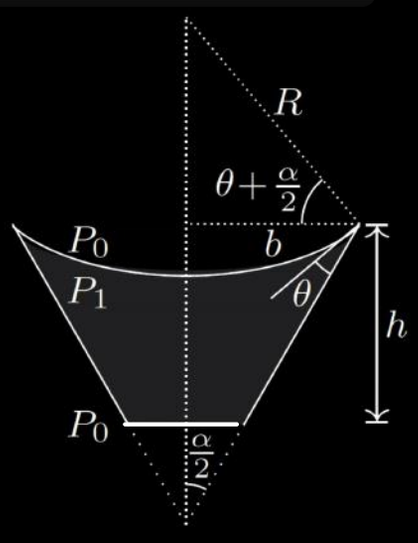
Sol: Refer to shown figure (Exaggerated)

$R = b\sec(\theta + \alpha/2)$

Now using Excess pressure logic:

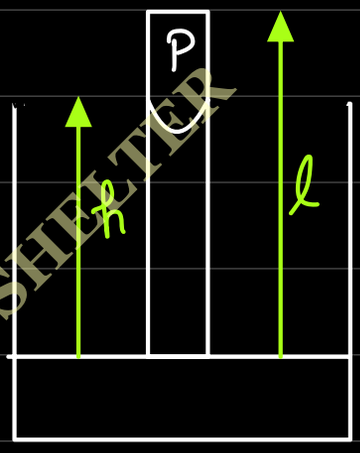
$$p_0 - \frac{2T}{R} + \rho g h = p_0 \Rightarrow h = 2T/R$$

$$\Rightarrow h = 2T\cos(\theta + \alpha/2)/b\rho g \quad \text{Ans.}$$



Example (Sealed Capillary): Irodov: 2·168

When a vertical capillary of length l with the sealed upper end was brought in contact with the surface of a liquid, the level of this liquid rose to the height h. The liquid density is ρ, the inside diameter of the capillary is d, the contact angle is θ, the atmospheric pressure is p_0. Find the surface tension of the liquid.



Sol. Assuming Isothermal Compression:

$$P \times \frac{\pi d^2}{4} \times (l-h) = p_0 \times l \times \frac{\pi d^2}{4} \quad (1)$$

$$\Rightarrow P(l-h) = p_0 l \quad (1)$$

Also using excess pressure:

$$P - \frac{2T}{d/2 \cos\theta} + \rho g h = p_0 \quad (2)$$

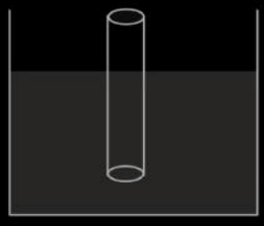
Solving (1) & (2) we get: $T = l \rho g h + p_0 \frac{l}{(l-h)} \cdot \frac{d}{4 \cos\theta}$

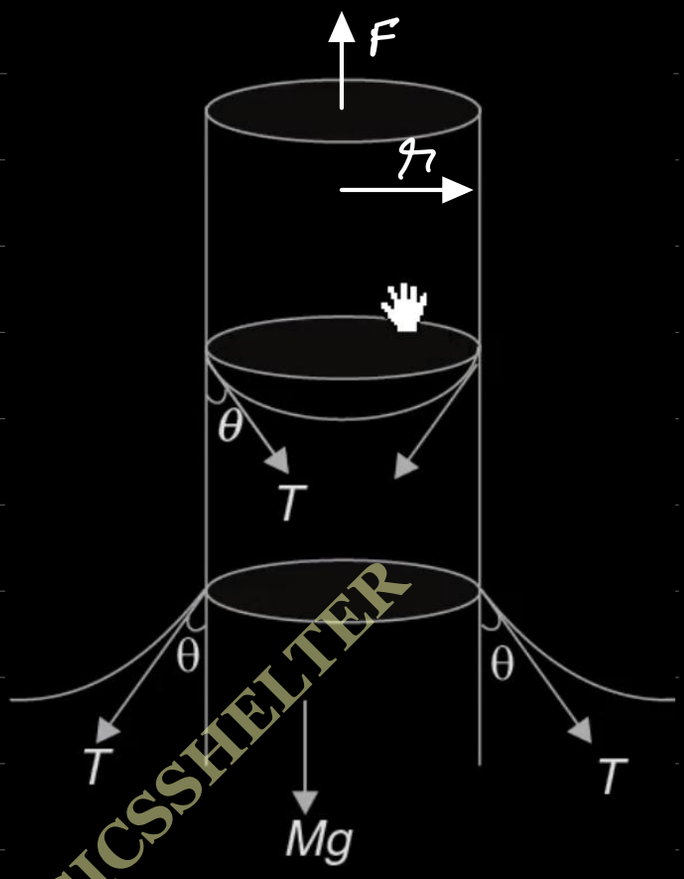
Ans.

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Example:

A long thin walled capillary tube of mass M and radius r is partially immersed in a liquid of surface tension T. The angle of contact for the liquid and the tube wall is 30°. How much force is needed to hold the tube vertically? Neglect buoyancy force on the tube.





Sol: Consider the FBD of the tube as shown:

$$2\pi r T\cos\theta + 2\pi r T\cos\theta + Mg = F$$

$$\Rightarrow F = Mg + 2\sqrt{3}\,\pi r T$$

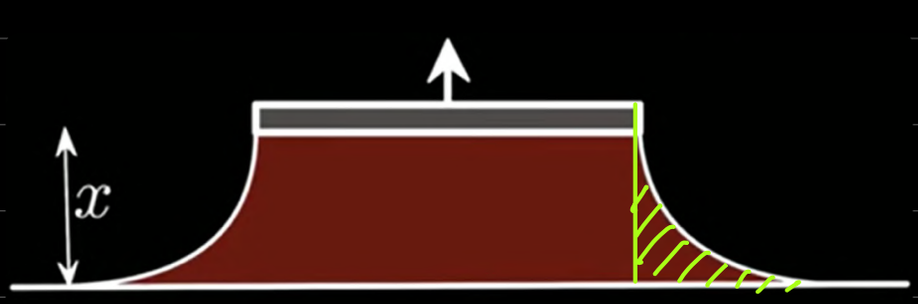
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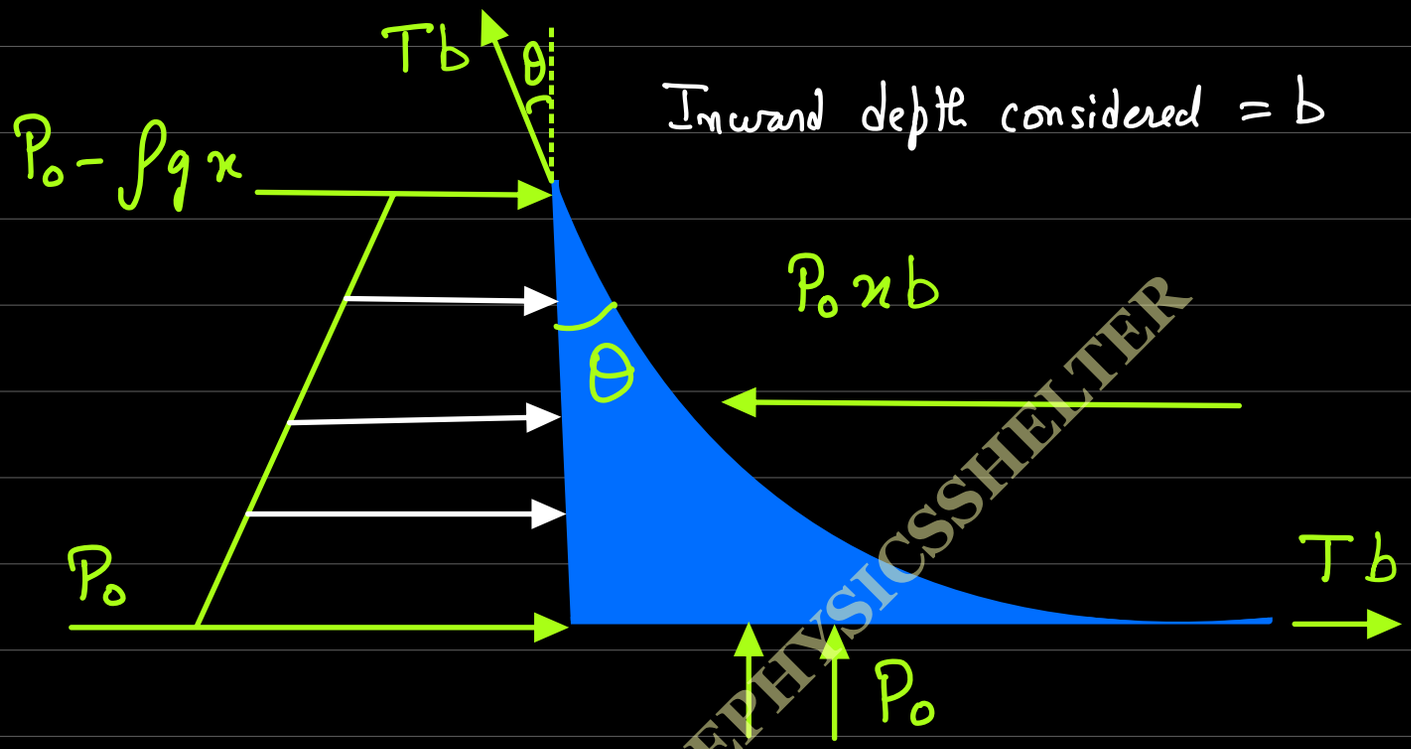
Example:

A long plate of width l is brought in contact with a liquid surface and then gradually pulled as shown in the figure. How does the required force per unit length vary with height x. Given liquid density is ρ and mass per unit length of the plate is λ.



Sol: Consider the FBD of the shaded portion of the liquid





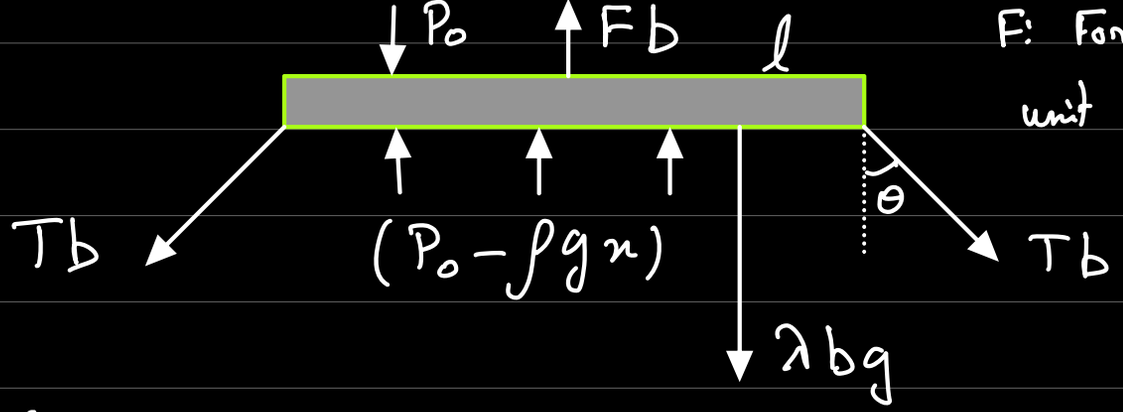
Inward depth considered = b

By horizontal Force Balance:

$$\left(P_0 - \rho g \frac{x}{2}\right)(xb) + Tb = Tb \sin\theta + P_0 x b$$

$$\Rightarrow \quad \sin\theta = 1 - (\rho g x^2 / 2T) \Rightarrow$$

Now Considering vertical Force Balance on the plate:



F: Force per unit length.

$$F_b + (P_0 - \rho g x) l b = 2 T_b \cos\theta + \lambda b g + P_0 l b$$

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$$\Rightarrow \quad F = \rho g x \ell + 2T\cos\theta + \lambda g$$

$$\Rightarrow \quad \boxed{F = \lambda g + \rho g x \left(\ell \sqrt{\frac{4T}{\rho g} - x^2} \right)} \qquad \text{Ans.}$$

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